Scan Architectures and Design Rules

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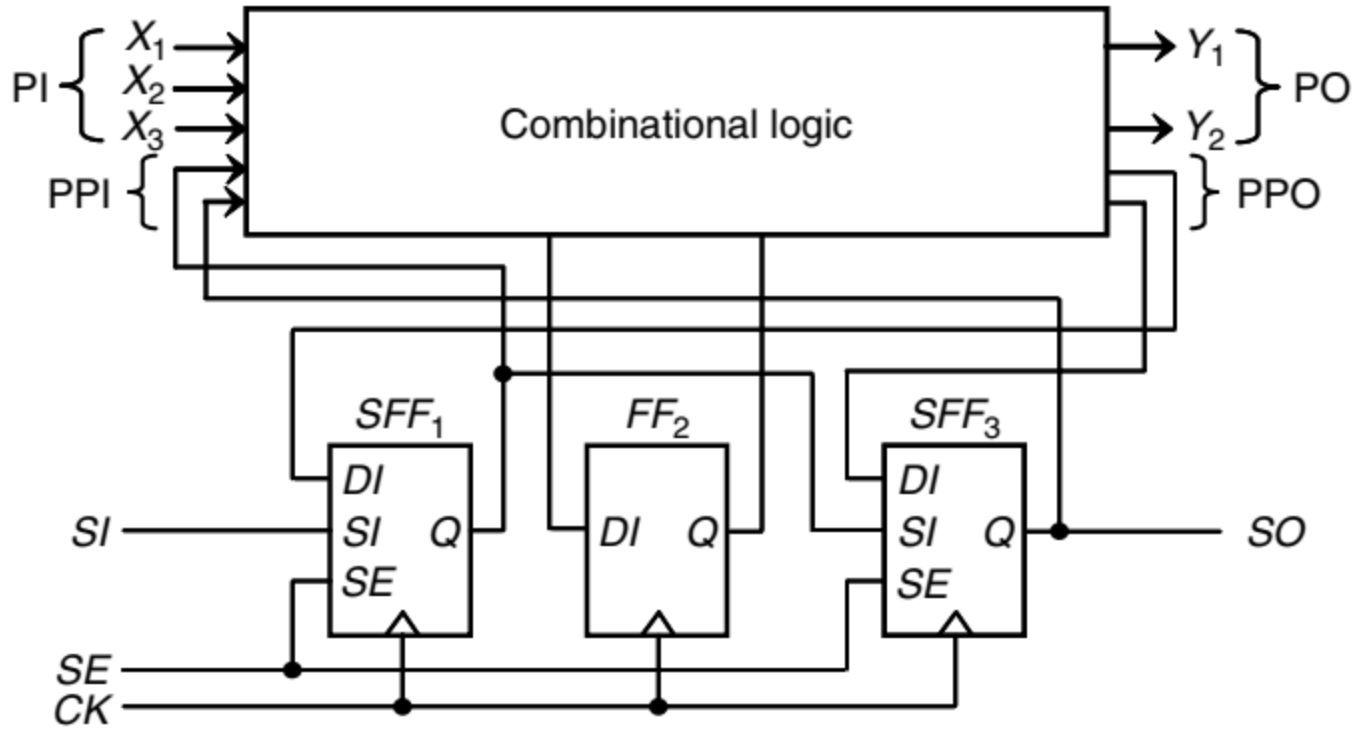
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Partial-Scan Design

- Partial-scan design **only requires** that a **subset of storage elements** be replaced with scan cells and connected into scan chains.
- **Sequential ATPG** has to be used to control and observe the value of the **non-scan flip-flop FF2** through SFF1 and SFF3.
- A subset of storage elements to be replaced with scan cells is selected to make the sequential circuit **feedback-free**.

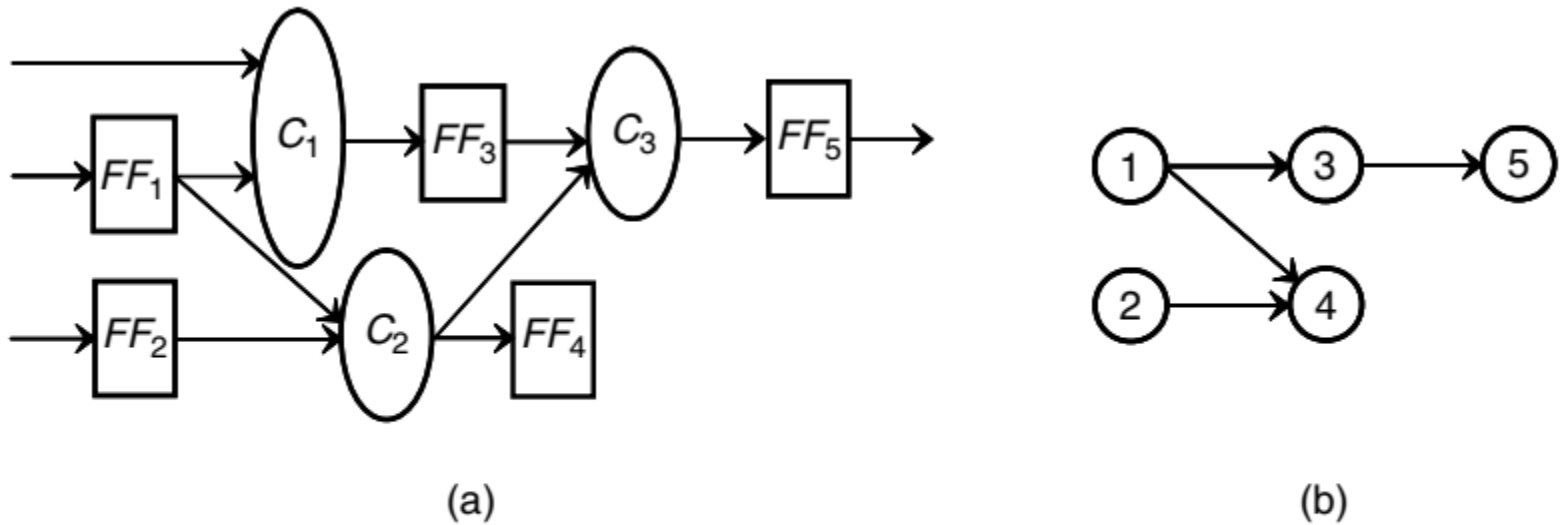


■ **FIGURE 2.17**

Partial-scan design.

Structure graph

- For a feedback-free sequential circuit, the structure graph is **a directed acyclic graph**.
- The **sequential depth** of a circuit is equal to the **maximum number of clock cycles** that must be applied in order to control and observe values to and from all non-scan storage elements.
- In a partial-scan design, **replacing a storage element with a scan cell** is equivalent to **removing its corresponding vertex** from the structure graph.



■ **FIGURE 2.18**

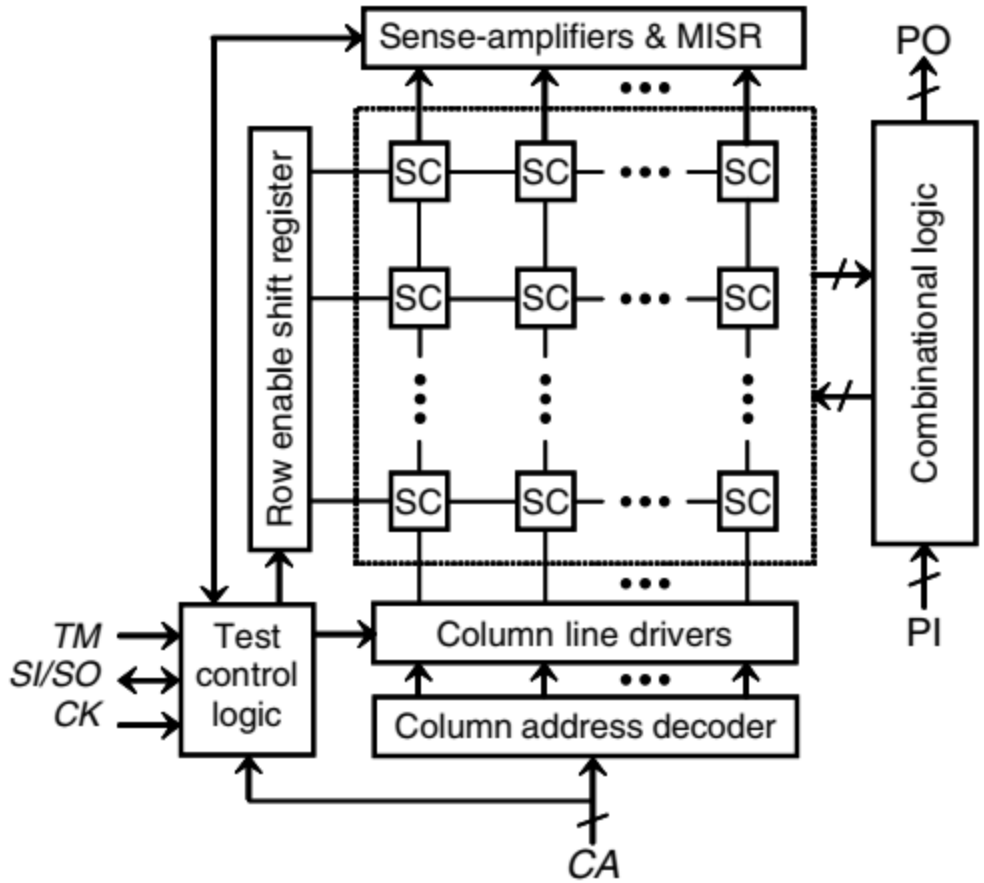
Sequential circuit and its structure graph: (a) sequential circuit, and (b) structure graph.

Balanced Partial-Scan

- In the balanced partial-scan design approach, a target **sequential depth** (eg, **3 to 5**) is used to further simplify the test generation process for the pipelined or feed-forward partial-scan design.
- By keeping the sequential depth under a small limit, one can apply **combinational ATPG using multiple time frames** to further increase the fault coverage of the design.

Random-Access Scan Design

- Random-access scan (RAS) makes **each scan cell** randomly and uniquely **addressable**, similar to storage cells in a random-access memory (RAM).
- The RAS design significantly **reduces test power dissipation** and simplifies the process of performing delay tests.



(b)

Progressive random-access scan (PRAS)

Progressive random-access scan (PRAS)

- The **row** enable shift register is used to **activate one row** of scan cells at a time.
- It is only necessary to **supply** a column address to specify which **scan cell** in an enabled row to access.
- The column address is provided in **parallel in one clock cycle** instead of providing a full address in multiple clock cycles.

Scan design rules

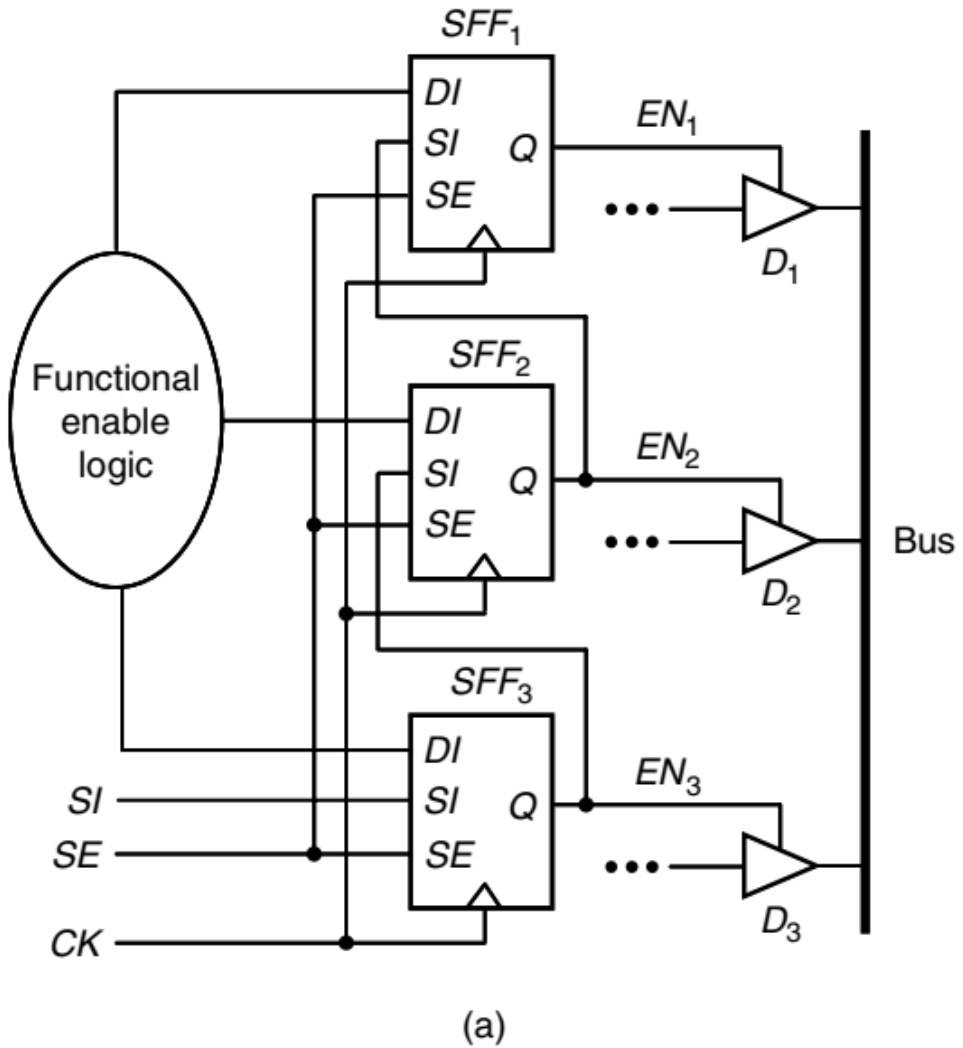
- In order to implement scan into a design, the design must comply with a set of **scan design rules**.
- In addition, a set of design styles must be avoided, as they may limit the fault coverage that can be achieved.

TABLE 2.7 ■ Typical Scan Design Rules

Design Style	Scan Design Rule	Recommended Solution
Tristate buses	Avoid during shift	Fix bus contention during shift
Bidirectional I/O ports	Avoid during shift	Force to input or output mode during shift
Gated clocks (muxed-D full-scan)	Avoid during shift	Enable clocks during shift
Derived clocks (muxed-D full-scan)	Avoid	Bypass clocks
Combinational feedback loops	Avoid	Break the loops
Asynchronous set/reset signals	Avoid	Use external pins

Tristate Buses

- Bus contention occurs when two bus drivers force opposite logic values onto a tristate bus.
- During the shift operation, no such guarantees can be made.
- Where EN1 is forced to 1 to enable the D1 bus driver, while EN2 and EN3 are set to 0 to disable both D2 and D3 bus drivers, when SE = 1.

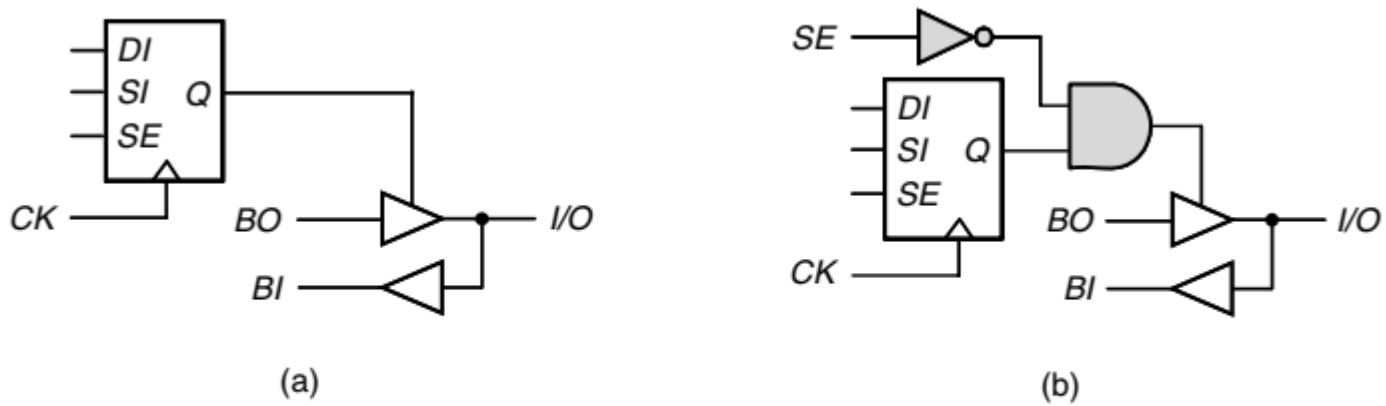


■ **FIGURE 2.21**

Fixing bus contention: (a) original circuit, and (b) modified circuit.

Bidirectional I/O Ports

- Conflicts may occur at a bidirectional I/O port during the shift operation.
- Because the output value of the scan cell can vary during the shift operation, the output tristate buffer may become active.
- Figure 2.22b shows an example of how to fix this problem by forcing the tristate buffer to be inactive when SE = 1.

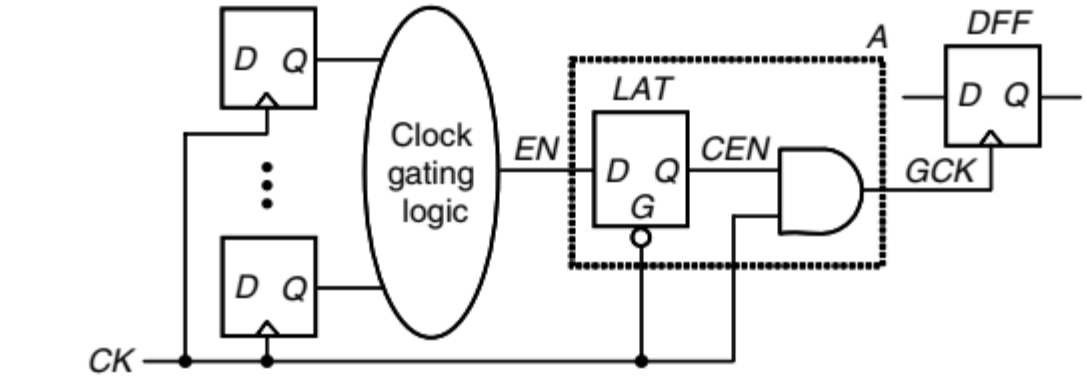


■ **FIGURE 2.22**

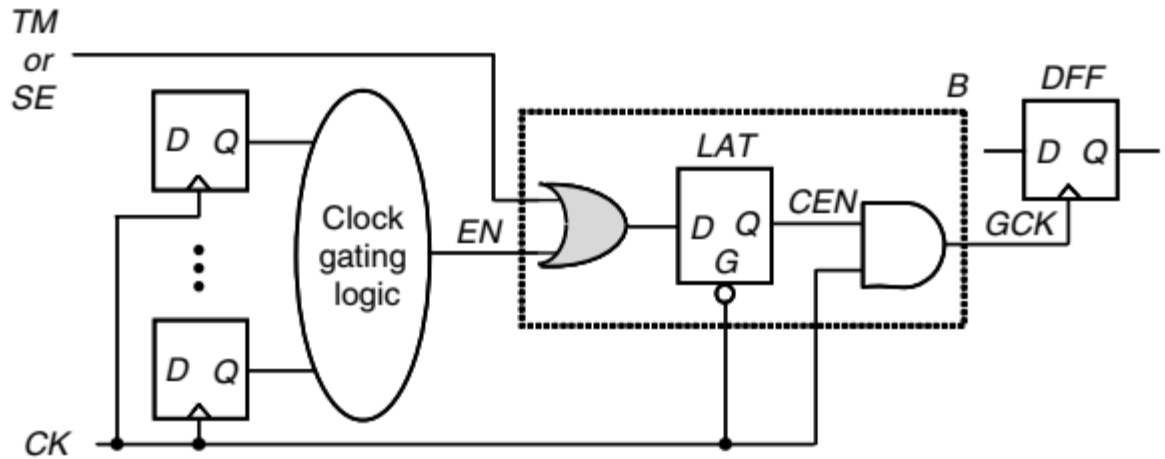
Fixing bidirectional I/O ports: (a) original circuit, and (b) modified circuit.

Gated Clocks

- Although clock gating is a good approach for reducing power consumption, it prevents the clock ports of some flip-flops from being directly controlled by primary inputs.
- As a result, modifications are necessary to allow the scan shift operation to be conducted on these storage elements.
- Figure 2.23b shows an example of how to **bypass the clock-gating logic during scan operation**.



(a)



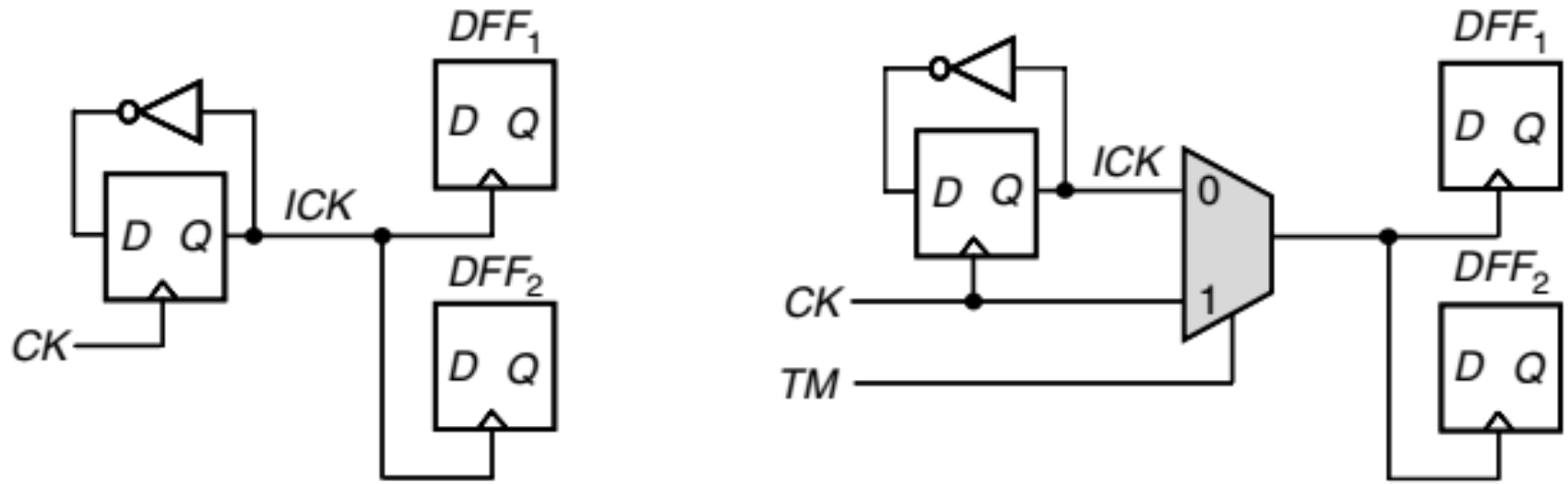
(b)

■ **FIGURE 2.23**

Fixing gated clocks: (a) original circuit, and (b) modified circuit.

Derived Clocks

- Because **derived clocks are not directly controllable** from primary inputs, in order to test the logic driven by these derived clocks, these clock signals must be bypassed during the entire test operation.
- Figure 2.24a shows an example where the derived clock ICK is generated internally and used to clock two storage elements.
- Figure 2.24b shows an example of how to **bypass the derived clock** during test operation.

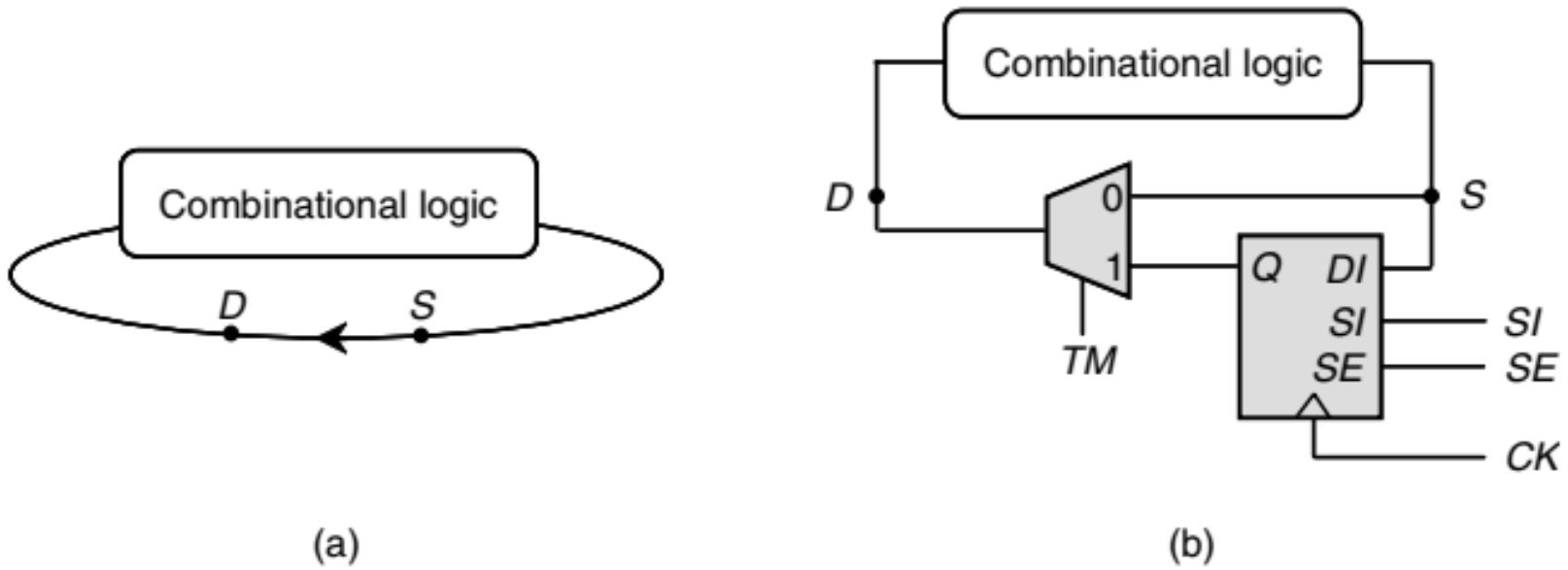


■ **FIGURE 2.24**

Fixing derived clocks: (a) original circuit, and (b) modified circuit.

Combinational Feedback Loops

- Depending on whether the **number of inversions** on a combinational feedback loop is **even or odd**, it can introduce either **sequential behavior or oscillation into a design**.
- Because the value stored in such a loop cannot be controlled or determined during test, it can increase test generation complexity or reduce fault coverage.
- Figure 2.25b shows how a **scan point** can be inserted to break the feedback loop during test mode.

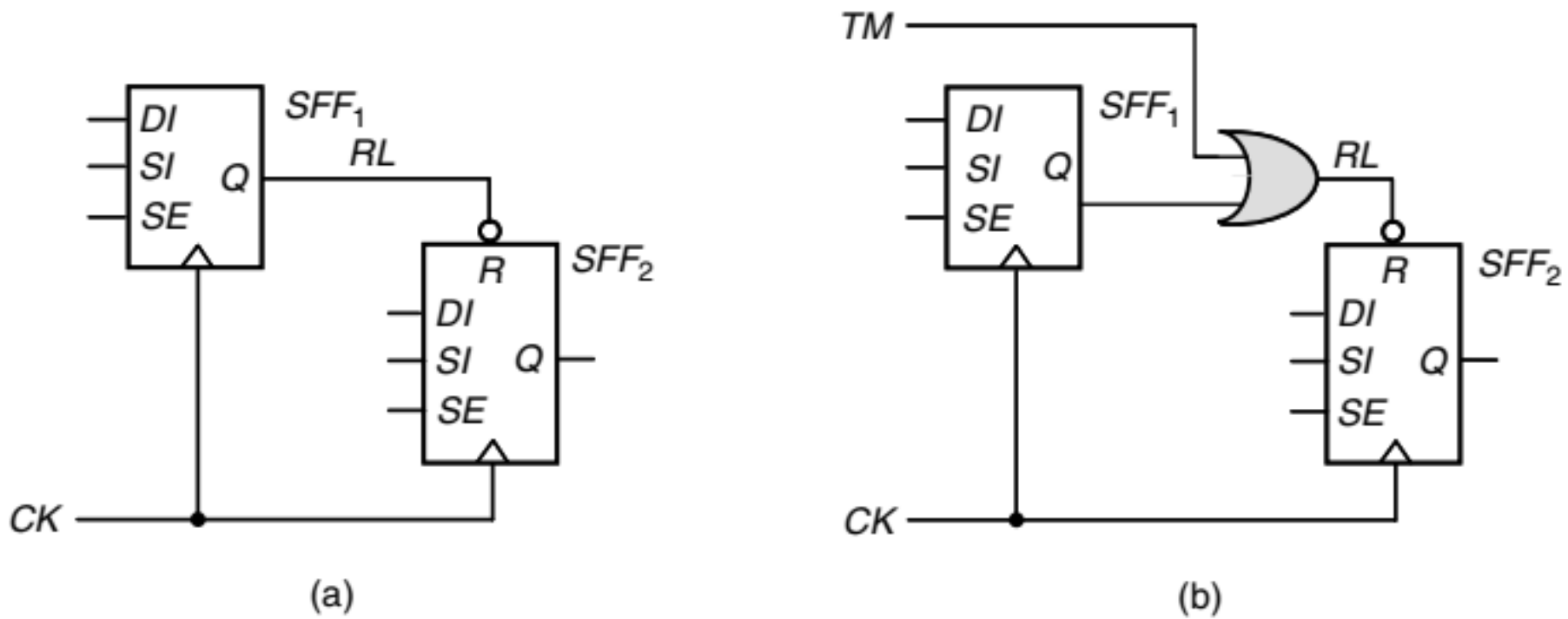


■ **FIGURE 2.25**

Fixing combinational feedback loops: (a) original circuit, and (b) modified circuit.

Asynchronous Set/Reset Signals

- Asynchronous set/reset signals can interfere with the scan shift operation if they are activated during test.
- During scan shifting, asynchronous set/reset signals should be **disabled.**
- Figure 2.26b shows an example of gating the asynchronous control signal with the scan enable signal.



■ **FIGURE 2.26**

Fixing asynchronous set/reset signals: (a) original circuit, and (b) modified circuit.